ML-SCODNet: Multitask Learning for Scene Classification and Object Detection Network from Remote Sensing Images *(PRCV2024)*

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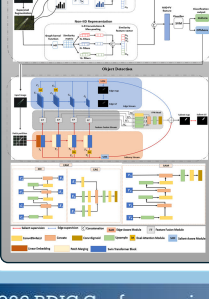
Abstract

This paper presents a novel and efficient multitask learning framework for scene classification and object detection from remote sensing images, which mainly contains the Non-Independent and Identically Distributed Fisher Vector Network (NIID-FVNet) for scene classification and Two-Stream CNN-Transformer Network (TSCTNet) for object detection.

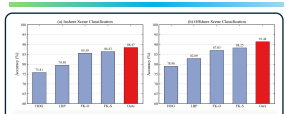


Methodology

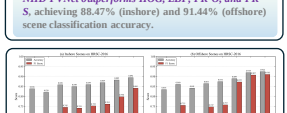
The method figure illustrates the overall ML-SCODNet architecture. NIID-FVNet first classifies an input remote sensing image as inshore or offshore by combining Fisher Vector encoding with non-IID graph-based features and an SVM classifier. The classified image is then fed into TSCTNet for object detection. TSCTNet uses a two-stream encoder: a CNN stream for edge details and a Swin Transformer stream for global salient features. A three-stream decoder—edge, saliency, and feature-fusion streams—together with dual-attention, edge-aware, and saliency-aware modules produces accurate ship detection results. The three decoder outputs are finally integrated.



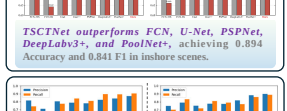
Experiments



***NIID-FVNet outperforms HOG, LBP, FK-Q, and FK-S,* achieving 88.47% (inshore) and 91.44% (offshore) scene classification accuracy.**



***TSCTNet outperforms FCN, U-Net, PSPNet, DeepLabv3+, and PoolNet+,* achieving 0.894 Accuracy and 0.841 F1 in inshore scenes.**



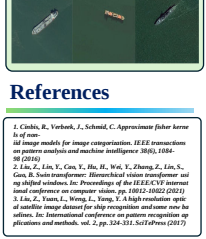
Our model outperforms others in precision and recall for both inshore and offshore scenes.



Visual results show our model achieves more accurate ship detection in inshore and offshore scenes.

Conclusions

- **Scene classification uses NIID-FVNet**
- **TSCTNet detects ships accurately.**
- **CNN-Transformer extracts features.**
- **Three-stream decoder enhances results.**
- **Future work reduces computation cost.**



References

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